

EMBRUN, France

WMO: 075910

Lat: 44.5656N

Lon: 6.5022E

Elev: 876

StdP: 91.23

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-8.4	-6.5	-17.6	0.9	-1.0	-15.4	1.1	-0.7	12.6	3.0	11.2	4.0	1.6	40	0.541

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.8	31.6	17.3	30.1	17.1	28.5	16.7	18.7	28.6	18.0	27.6	17.3	26.4	4.6	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.4	12.2	20.9	14.6	11.5	20.6	13.8	11.0	20.3	56.6	28.7	54.1	27.5	52.0	26.5	21.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
10.4	9.0	7.6	DB	-10.7	33.7	2.2	1.6	-12.3	34.9	-13.6	35.8	-14.9	36.7	-16.5	37.8	(4)
			WB	-11.8	20.0	2.2	0.7	-13.4	20.5	-14.7	20.9	-16.0	21.3	-17.6	21.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	11.0	2.0	3.2	7.1	10.0	13.9	18.2	20.5	20.2	15.9	12.0	6.1	2.7
	DBStd	7.38	3.07	3.74	3.65	3.58	3.41	3.55	3.15	3.21	3.31	3.38	3.66	3.46
	HDD10.0	963	247	192	104	44	7	0	0	0	2	19	123	226
	HDD18.3	2883	505	424	349	251	142	47	16	17	86	196	366	484
	CDD10.0	1340	0	1	13	43	128	245	326	317	179	82	7	0
	CDD18.3	219	0	0	0	0	5	42	83	76	13	0	0	0
	CDH23.3	2456	0	0	0	8	69	471	921	820	159	7	0	0
	CDH26.7	771	0	0	0	1	9	135	317	286	24	0	0	0
Wind	WSAvg	3.0	2.8	3.0	3.3	3.3	3.1	3.0	3.0	2.8	2.7	2.8	2.9	2.9
Precipitation	PrecAvg	824	59	47	49	71	71	71	55	56	79	101	100	64
	PrecMax	1052	127	134	234	171	173	150	122	96	249	221	258	181
	PrecMin	538	1	0	1	15	20	22	19	18	12	8	18	0
	PrecStd	134	37	36	47	38	33	30	27	20	51	55	71	44
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.0	17.0	21.1	24.4	27.6	32.3	33.0	33.6	29.1	24.4	17.7	13.1
		MCWB	5.5	7.3	9.6	12.3	15.7	17.8	17.7	17.8	16.5	13.7	9.9	5.9
	2%	DB	10.3	13.9	18.7	21.7	25.2	30.1	31.6	31.6	27.0	22.0	15.5	11.1
		MCWB	4.6	5.9	8.6	11.2	14.4	17.3	17.4	17.0	15.8	13.2	8.8	5.0
	5%	DB	8.5	11.4	16.6	19.6	23.3	28.0	30.1	29.9	25.2	20.0	13.5	9.3
		MCWB	3.6	4.8	7.7	10.2	13.5	16.7	17.3	16.9	14.8	12.5	7.7	4.3
	10%	DB	6.8	9.2	14.3	17.5	21.3	26.1	28.4	28.0	23.2	18.2	11.6	7.7
		MCWB	2.6	3.6	6.7	9.2	12.6	16.0	16.8	16.5	14.1	11.6	7.0	3.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.1	8.1	10.6	12.8	16.6	19.7	19.7	19.3	17.7	15.5	11.0	7.8
		MCDB	10.7	15.5	17.9	22.0	25.9	29.1	30.0	29.6	26.1	21.3	15.1	10.0
	2%	WB	5.4	6.6	9.4	11.7	15.2	18.2	18.7	18.4	16.5	14.3	9.7	6.4
		MCDB	8.9	12.2	16.8	20.4	23.5	27.7	28.7	28.4	25.1	19.9	13.4	9.1
	5%	WB	4.3	5.3	8.3	10.7	14.2	17.3	18.0	17.7	15.7	13.2	8.6	5.3
		MCDB	7.4	10.3	15.4	18.4	21.7	26.1	27.7	27.2	23.4	18.5	12.4	8.1
	10%	WB	3.2	4.2	7.2	9.7	13.2	16.5	17.3	17.0	14.7	12.2	7.6	4.2
		MCDB	6.1	8.5	13.4	16.4	20.0	24.7	26.6	25.8	21.6	16.9	11.0	7.1
Mean Daily Temperature Range	5% DB	MDBR	9.0	10.6	11.8	11.9	12.2	13.3	13.8	13.8	12.7	11.1	9.3	8.5
		MCDBR	10.9	13.9	15.2	15.0	14.8	15.2	15.5	15.4	14.9	13.1	11.6	10.3
		MCWBR	6.6	8.0	7.8	7.1	6.5	6.0	5.5	5.4	6.2	6.7	6.7	6.2
	5% WB	MCDBR	9.6	12.5	13.8	14.1	13.4	14.1	14.2	13.8	13.5	11.7	10.1	8.4
MCWBR		6.2	7.6	7.4	6.9	6.3	6.0	5.5	5.4	5.9	6.1	6.2	5.5	
Clear-Sky Solar Irradiance	taub	0.258	0.283	0.328	0.370	0.383	0.388	0.377	0.367	0.352	0.331	0.293	0.260	
	taud	2.408	2.371	2.339	2.284	2.308	2.330	2.361	2.395	2.422	2.473	2.497	2.448	
	Ebn at Noon	878	911	907	890	887	880	886	885	870	844	827	841	
	Edh at Noon	72	91	108	125	126	124	119	111	99	83	67	63	
All-Sky Solar Radiation	RadAvg	1.90	2.84	3.93	4.84	5.52	6.47	6.70	5.84	4.51	2.98	1.86	1.62	
	RadStd	0.25	0.32	0.50	0.51	0.47	0.40	0.45	0.32	0.36	0.34	0.29	0.18	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		+0.39	N/A	N/A	+0.94	N/A	N/A	N/A	N/A	N/A	+60
Regional (46 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page